

Final Project – D4E113

Life Expectancy

*Statistical Analysis on factors influencing Life Expectancy between
Developed & Developing countries*



Nhóm 3

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Agenda

Overview

Analysis

Conclusion

Overview

- The research examines how **health, immunization, and lifestyle factors** affect **life expectancy** worldwide **over the period of 2000 – 2015**.
- It also considers the impact of **mortality rates** and **socio-economic conditions** on life expectancy.
- With **sources** from WHO Global Health Observatory & United Nations

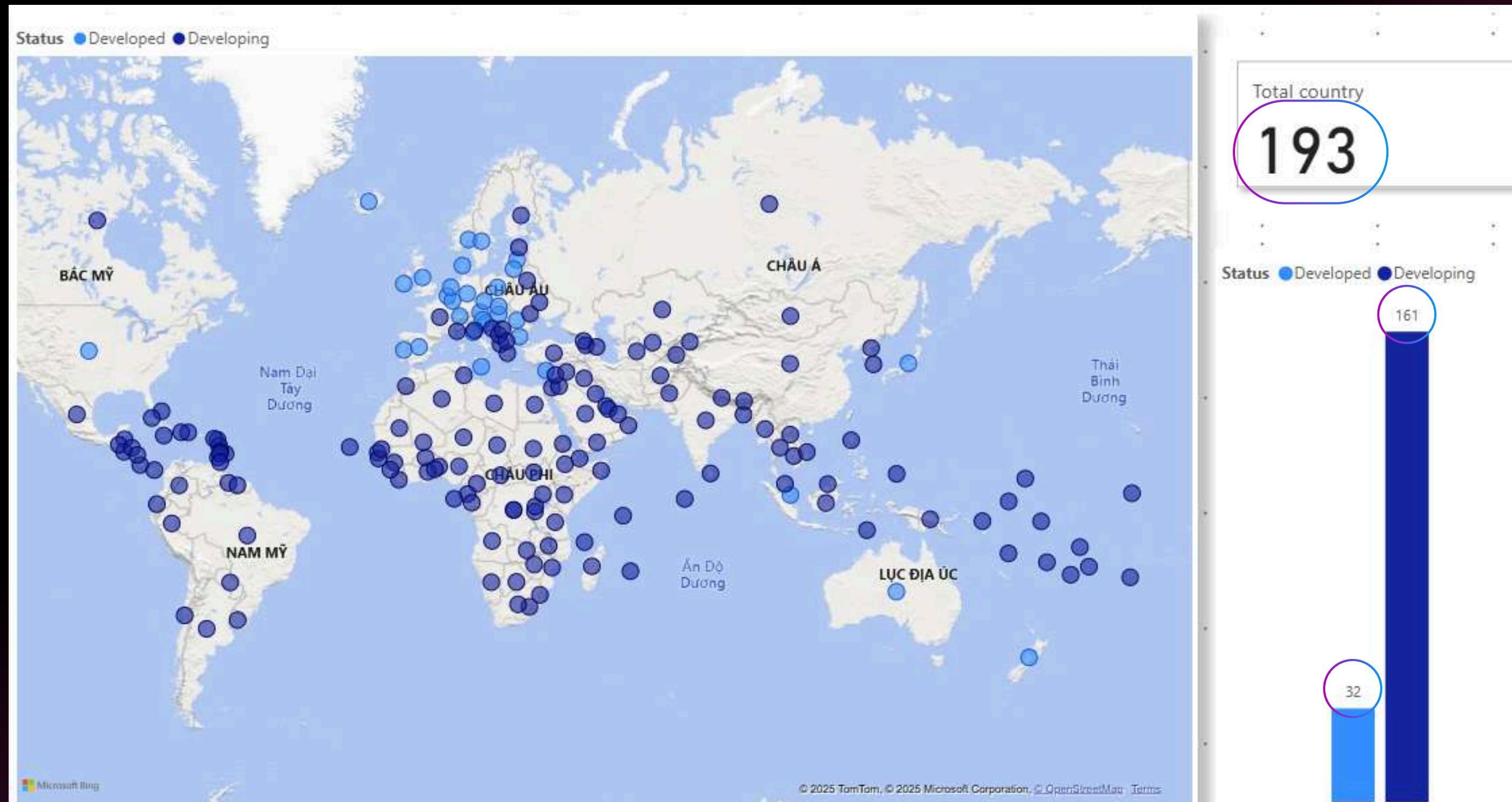


Figure 1. Countries distribution & status of dataset

About the Dataset

Includes **22 variables**, grouped into the following categories:

Identification & Demographics

Country
Year
Status

Lifestyle & Health Behavior

Alcohol
BMI
Thinness 1–19 years
Thinness 5–9 years

Immunization-Related Factors

Hepatitis B
Polio
Diphtheria
Measles

Life Expectancy Target Variable

Life Expectancy

Economic Factors

Percentage expenditure
Total expenditure
GDP
Income composition of resources

The Tools

01



→ **Clean** and **prepare** the raw dataset

02



→ **Visualization** of the prepared dataset for **insight generation**

Preprocessing Data



01

Clean column names – replace unwanted spaces with the underscore

```
df.columns = df.columns.str.strip().str.replace(" ", "_").str.replace("-", "_")
```

02

Dataset review – rows, columns, number of countries and timespan

```
print(f"Number of column and row of dataset: {df.shape}")  
print(f"Number of countries covered: {df['Country'].nunique()}")  
print(f"Years covered in the timespan: {df['Year'].min()} - {df['Year'].max()}")
```

Number of column and row of dataset: (2938, 25)
Number of countries covered: 193
Years covered in the timespan: 2000 - 2015

03 Inspect & Handle missing values

Step 4: Checking for missing values

```
missing = df.isnull().sum()
missing = missing[missing > 0].sort_values(ascending=False)
print("Missing values per column:\n", missing)
```

🌸 Missing values per column:

Population	652
Hepatitis_B	553
GDP	448
Total_expenditure	226
Alcohol	194
Income_composition_of_resources	167
Schooling	163
thinness_5_9_years	34
thinness_1_19_years	34
BMI	34
Polio	19
Diphtheria	19
Life_expectancy	10
Adult_Mortality	10
dtype: int64	

Step 5: Fill missing with column means

```
df.fillna(df.mean(numeric_only=True), inplace=True)
```


04

Add in columns and values – define categories for Life expectancy, Average rate of Immunization, Alcohol Consumption Levels

```
# Step 6: Create tables for low life expectancy, immunization and alcohol
df['Is_LE_Low'] = df['Life_expectancy'] < 65
df['Avg_Immunization'] = df[['Hepatitis_B', 'Polio', 'Diphtheria']].mean(axis=1)
df['Alcohol_Level'] = pd.cut(df['Alcohol'], bins=[-0.1, 2, 6, df['Alcohol'].max()],
                             labels=['Low', 'Moderate', 'High'])
```

Generate overview charts of Life Expectancy distribution & Development Status

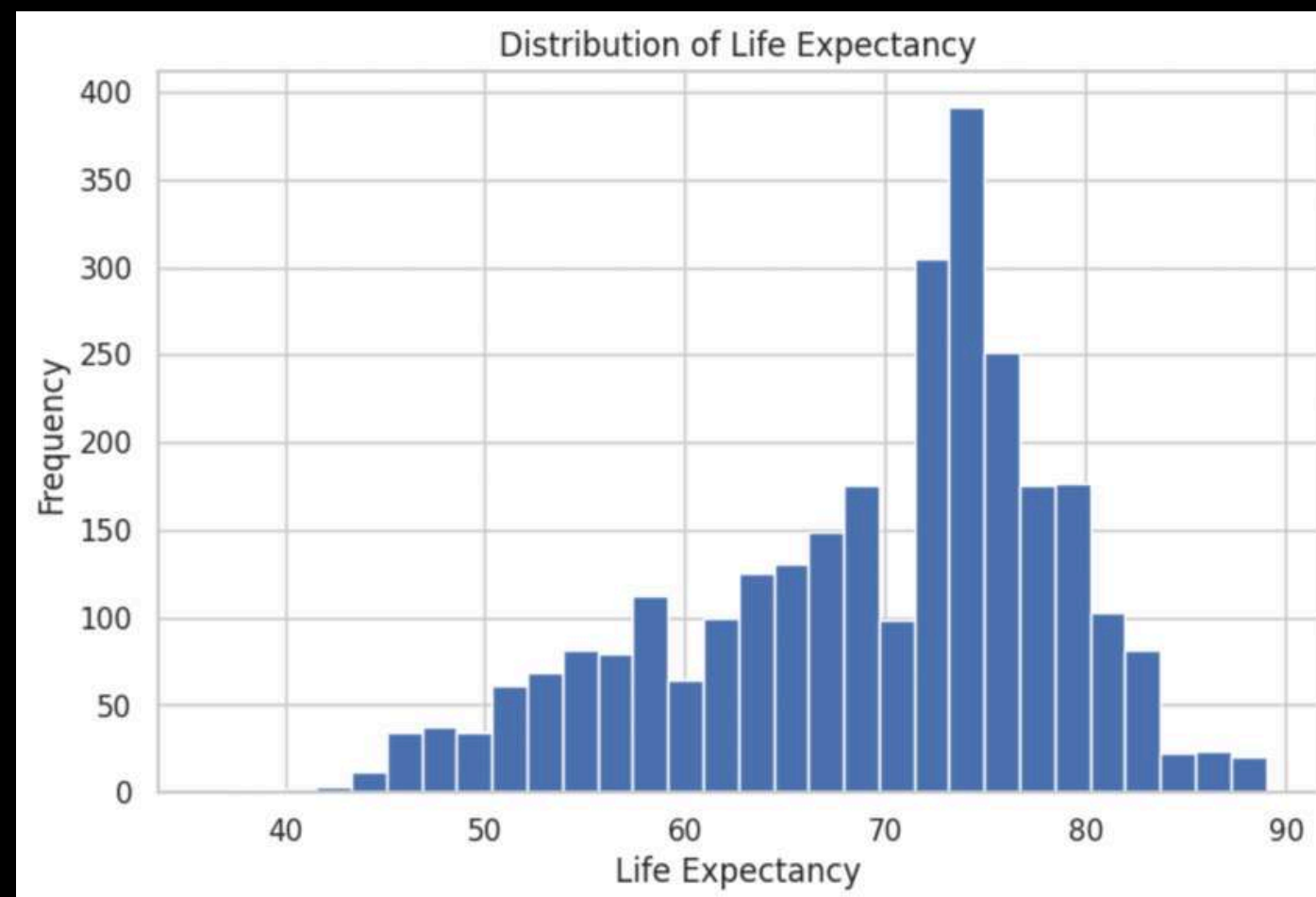


Figure 2. Distribution of Life Expectancy

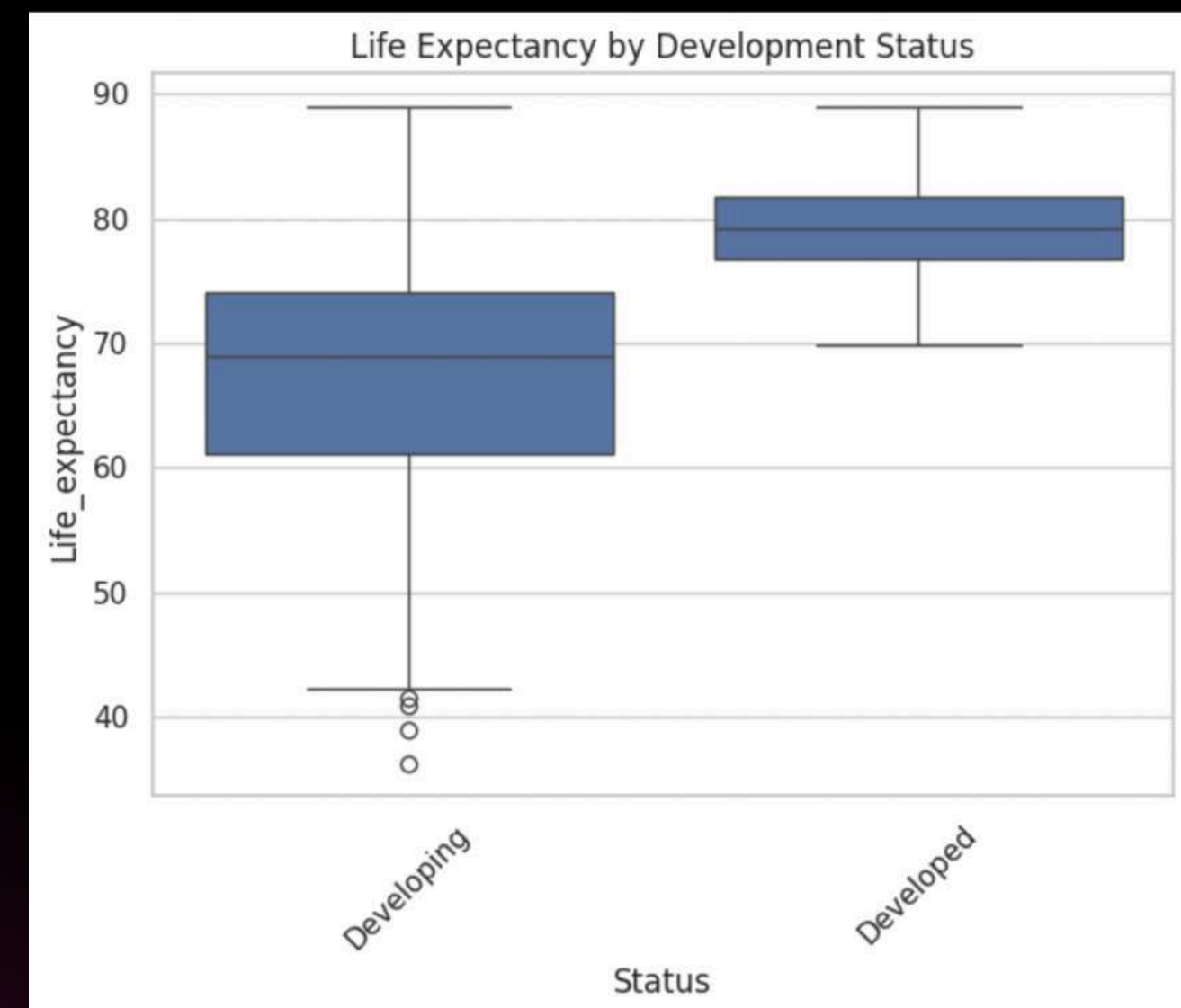


Figure 3. Average Life Expectancy by Development Status

Step 8: Correlation Analysis

```
corr_matrix = df.corr(numeric_only=True)
life_corr = corr_matrix['Life_expectancy'].sort_values(ascending=False)
print("Top positive correlations:\n", life_corr.head(10))
print("Top negative correlations:\n", life_corr.tail(5))
```

Top positive correlations:

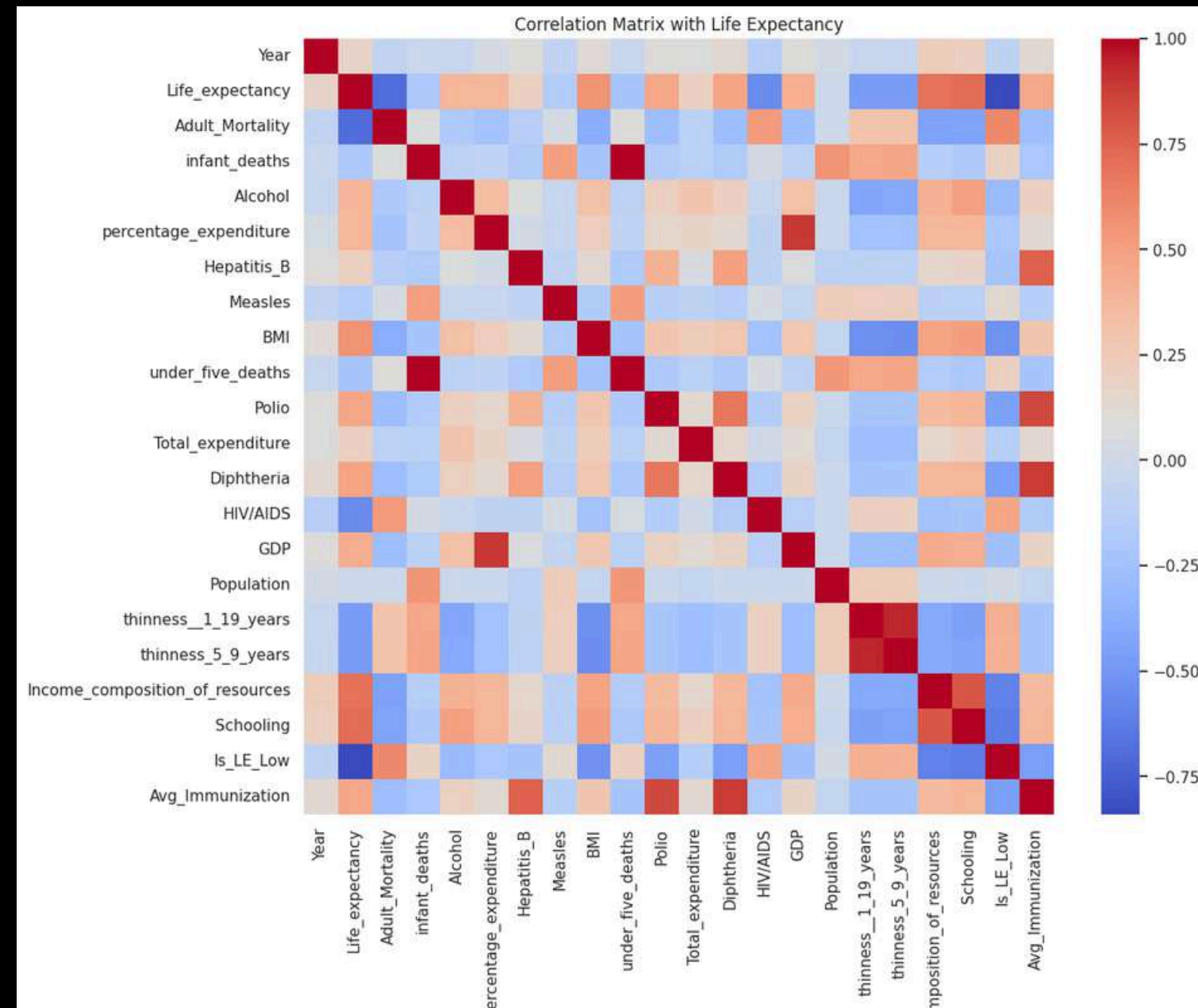
Life_expectancy	1.000000
Schooling	0.715066
Income_composition_of_resources	0.692483
BMI	0.559255
Diphtheria	0.475418
Avg_Immunization	0.461801
Polio	0.461574
GDP	0.430493
Alcohol	0.391598
percentage_expenditure	0.381791

Name: Life_expectancy, dtype: float64

Top negative correlations:

thinness_5_9_years	-0.466629
thinness_1_19_years	-0.472162
HIV/AIDS	-0.556457
Adult_Mortality	-0.696359
Is_LE_Low	-0.840149

Name: Life_expectancy, dtype: float64



Heatmap

```
plt.figure(figsize=(12, 10))
sns.heatmap(corr_matrix, cmap='coolwarm', annot=False)
plt.title("Correlation Matrix with Life Expectancy")
plt.show()
```

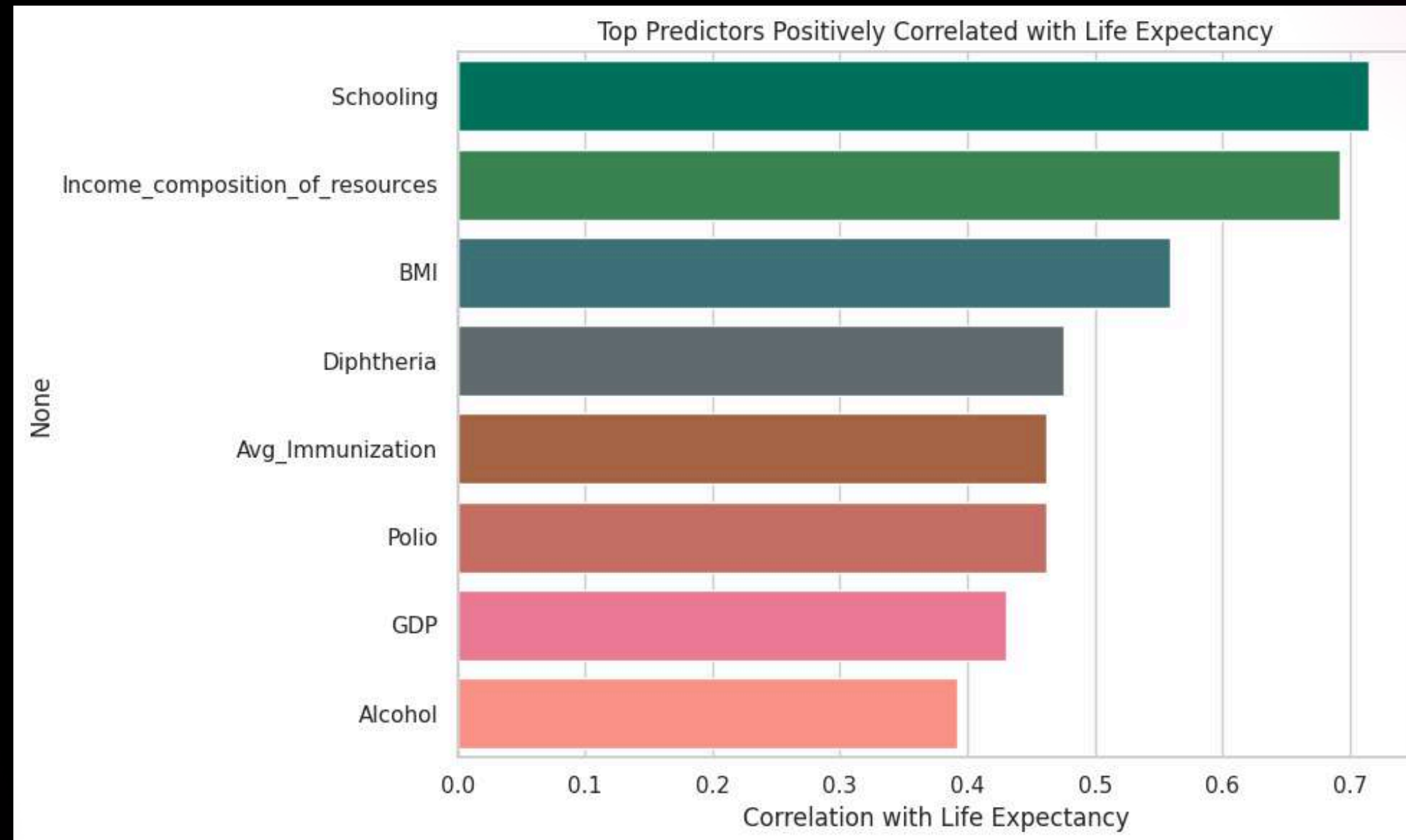
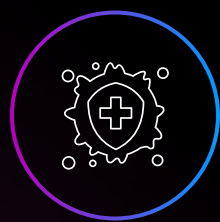


Figure 4. Top factors positively correlated with Life Expectancy

Analysis

Focus questions



Question 1

What is the impact of
Immunization coverage on life
expectancy?



Question 2

Does Life Expectancy has positive
or negative correlation with
**eating habits, lifestyle, exercise,
smoking, drinking alcohol, etc.?**

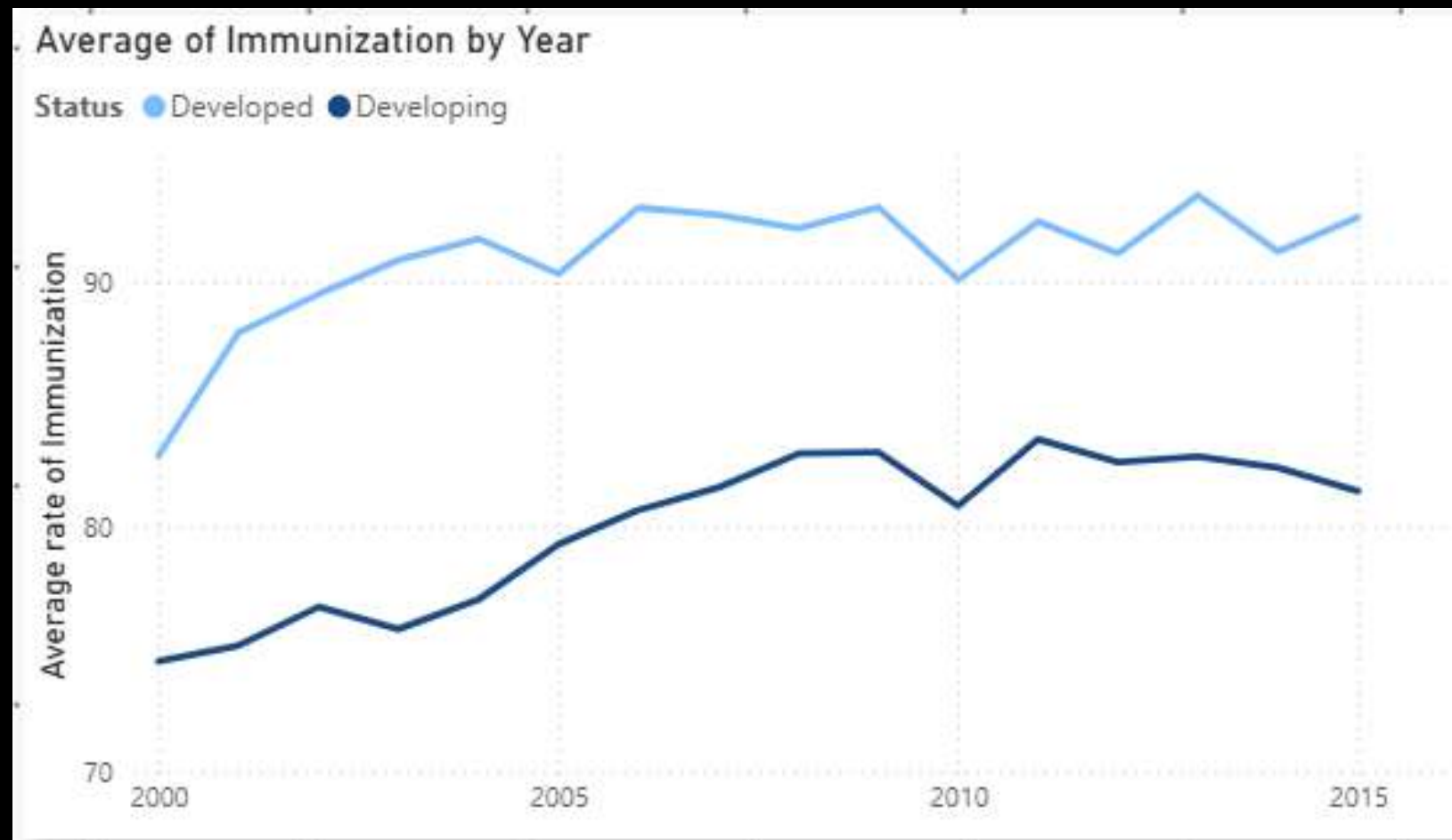


Figure 5. Average of Immunization Coverage by Year

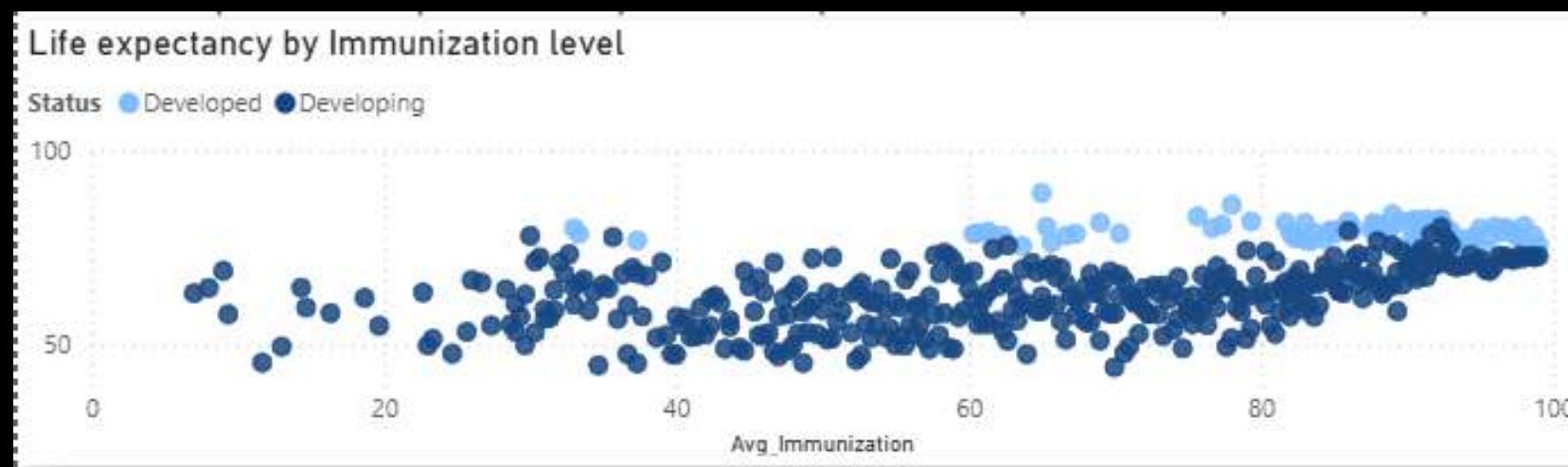


Figure 6. Average Life Expectancy compared to Immunization Coverage

- Developed countries have **strong vaccination programs**.
- Vaccines help **reduce child deaths** and **increase life expectancy**.
- To improve life expectancy further, social and economic changes are also needed.

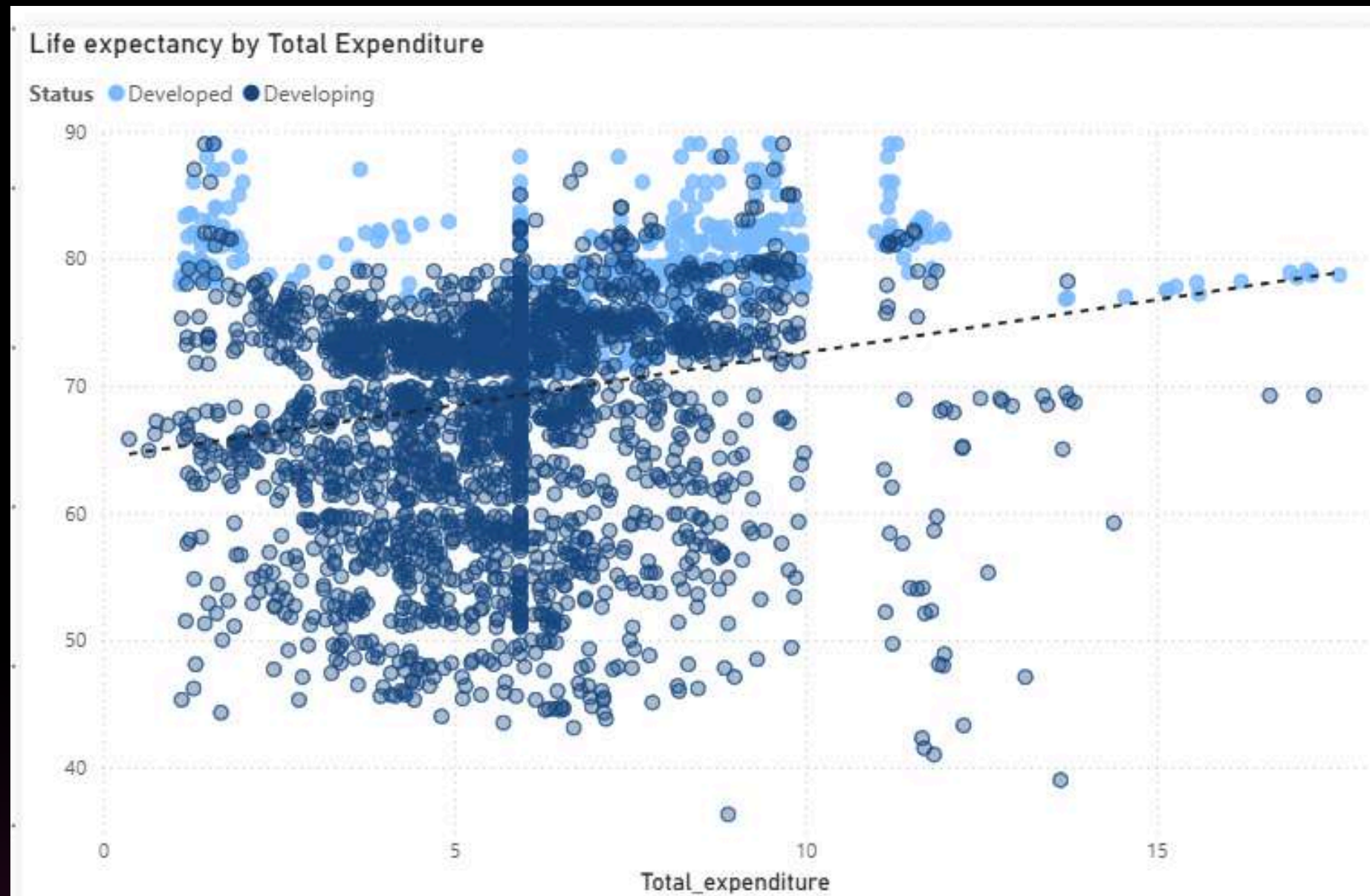
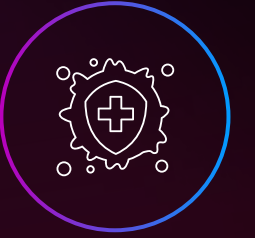


Figure 7. Average Life Expectancy compared to **Total Expenditure** on health

-
- Developed countries have **higher life expectancy** even with **moderate spending**.
 - Developing countries show **more variation**, suggesting lifestyle and other factors matter too.

**Total expenditure: General government expenditure on health as a percentage of total government expenditure (%)*

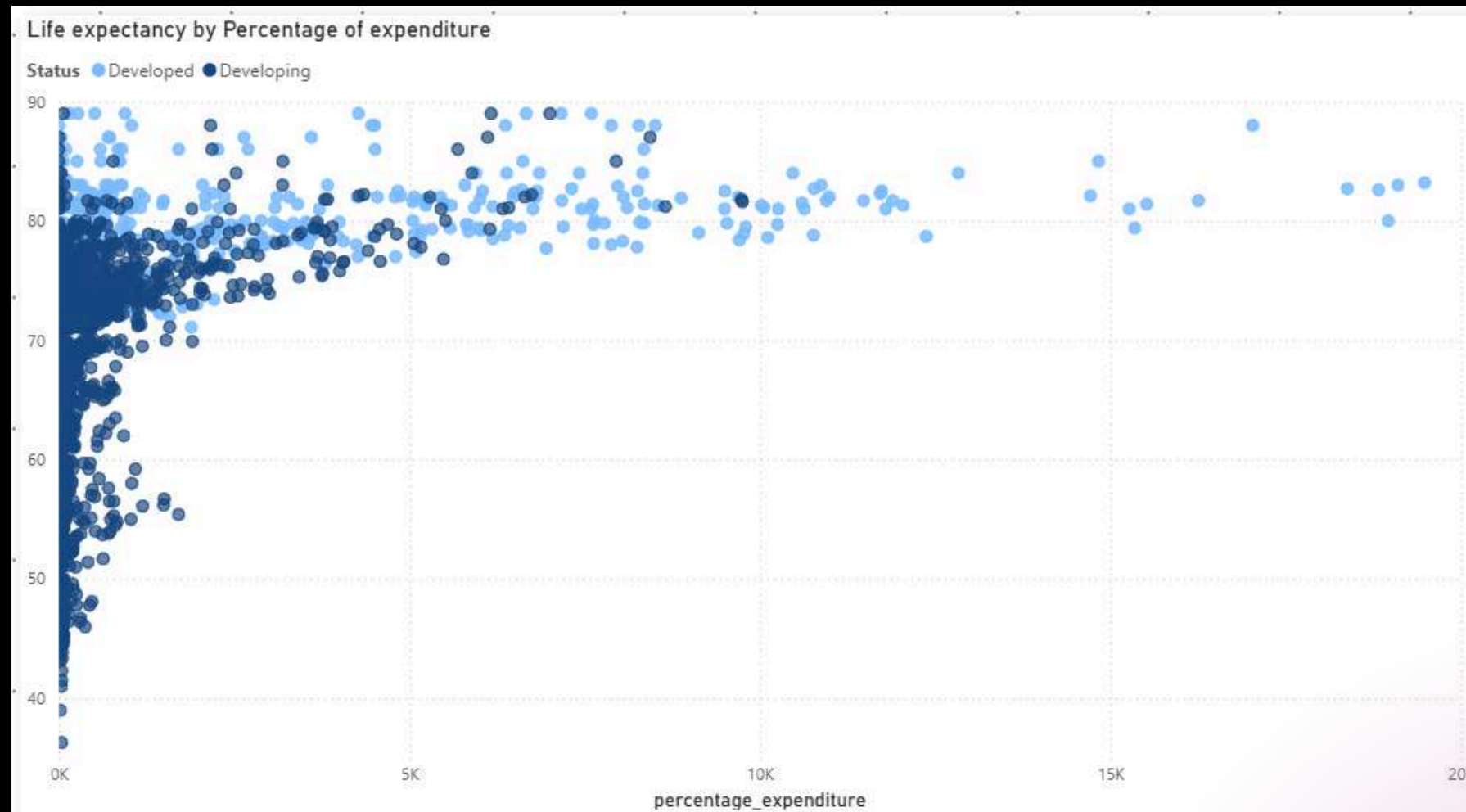
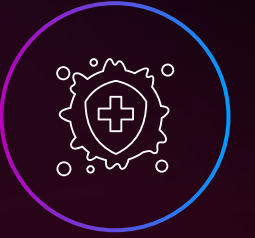
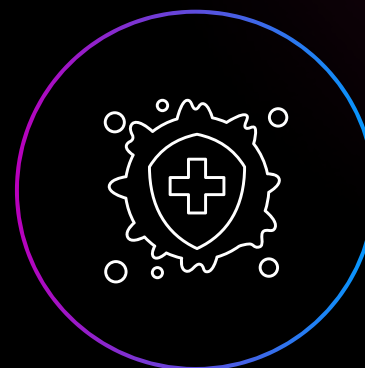


Figure 8. Average Life Expectancy compared to **Percentage of expenditure** on health

Developed countries have **high life expectancy** regardless of **differences in spending**.

→ Suggests healthcare quality, policies, and lifestyle are also important factors.

**Percentage of expenditure: Expenditure on health as a percentage of Gross Domestic Product per capita(%)*

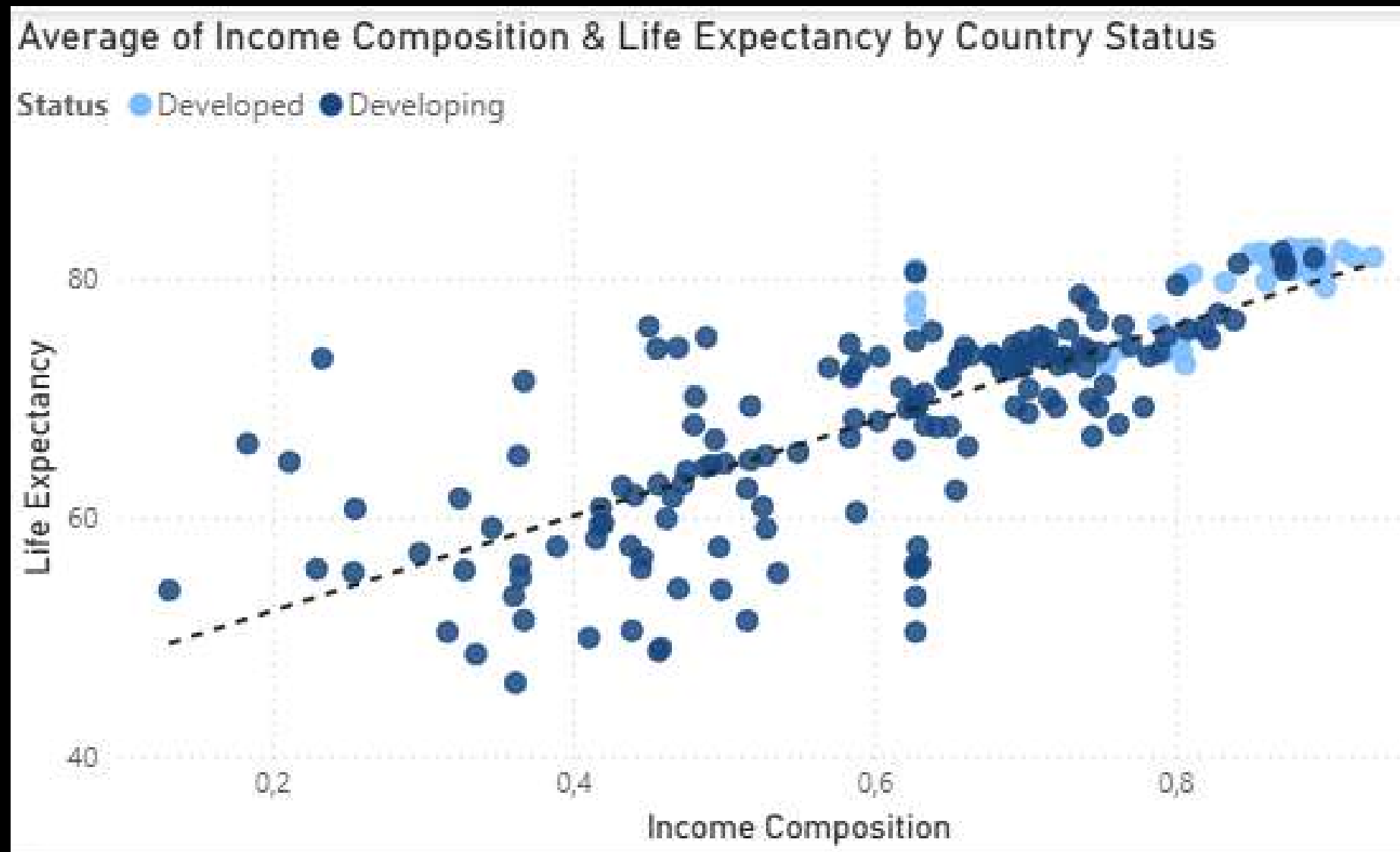


Immunization coverage demonstrates a **clear positive impact** on life expectancy across countries.

- **Consistent** across different regions and income levels
- Immunization is especially **valuable** in developing countries with weaker healthcare systems.
- While healthcare spending also helps, immunization has a **more direct and consistent impact**.



Investing in vaccination programs is a **cost-effective** way to improve public health and extend lifespan, particularly in low- and middle-income nations.

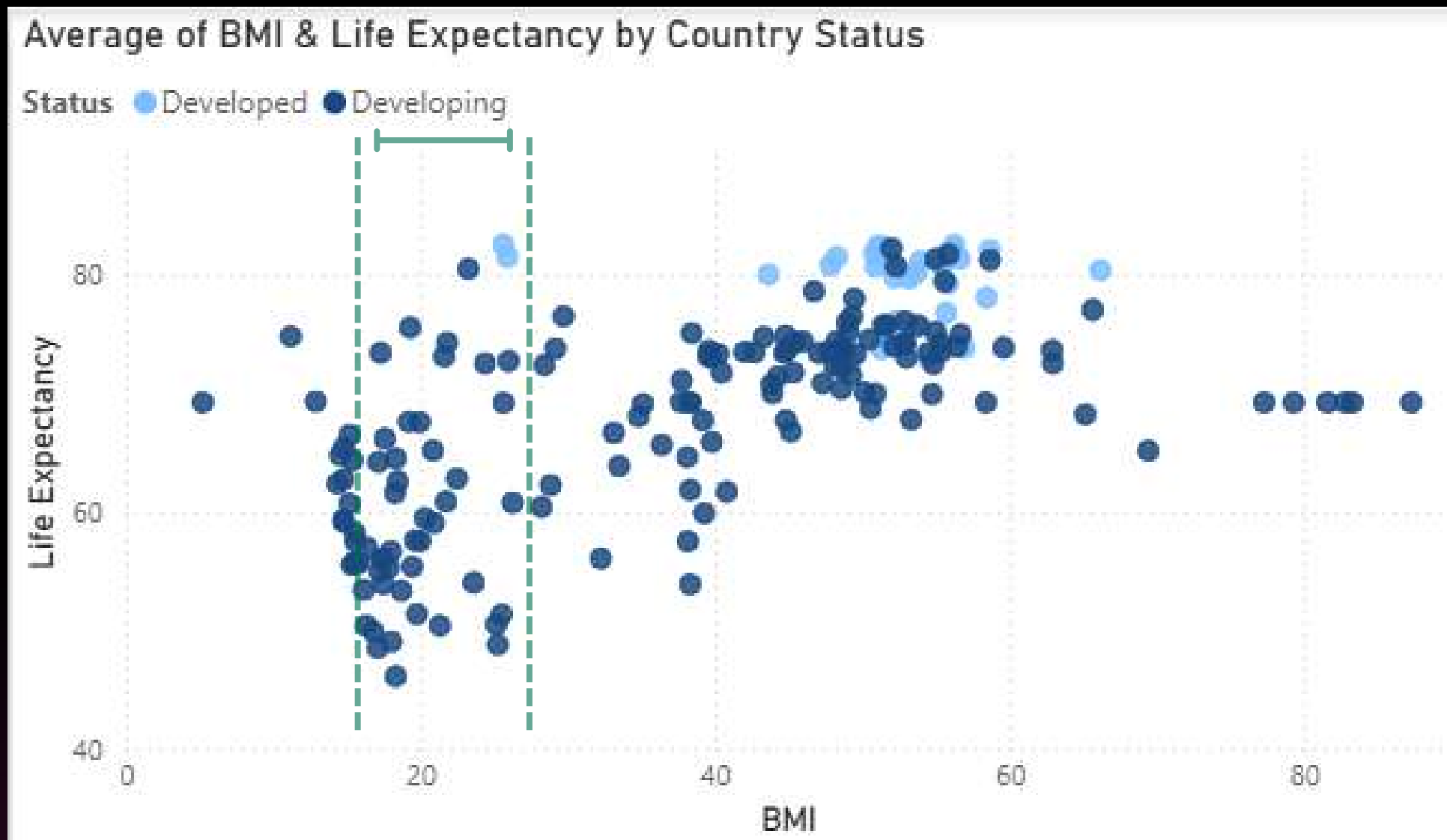


- Demonstrates **strong positive correlations**.
- Developed countries cluster in the top right.



Higher income leads to higher life expectancy

Figure 9. Average of Income Composition compared to Average of Life Expectancy



- **Healthy** BMI index: ~18.5 – 24.9 (NIH,n.d.)
- Very low BMI (underweight) is **more frequent** in developing countries.
- Higher BMI in developed countries does not always lead to lower life expectancy.



Moderate correlations

Figure 10. Average of BMI compared to Average of Life Expectancy

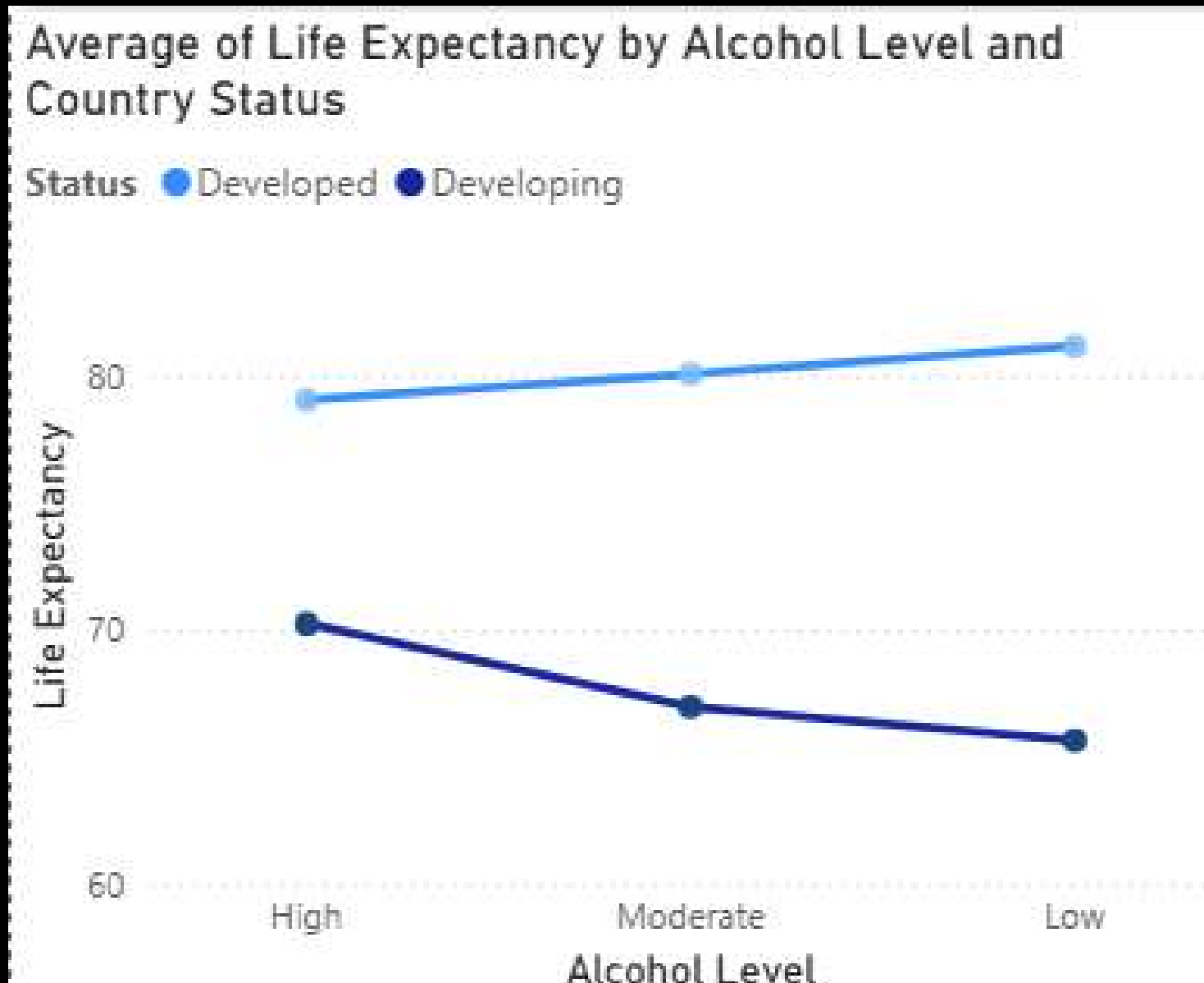


Figure 11. Average of Life Expectancy by Alcohol Level Usage

- Developed countries: Life Expectancy *slightly increases* as alcohol consumption decreases.



Positive correlation

- Developing countries: Life Expectancy *drops* as alcohol consumption decreases.



Negative correlation

**May consider other factors: reporting accuracy, other socioeconomic variables,...*

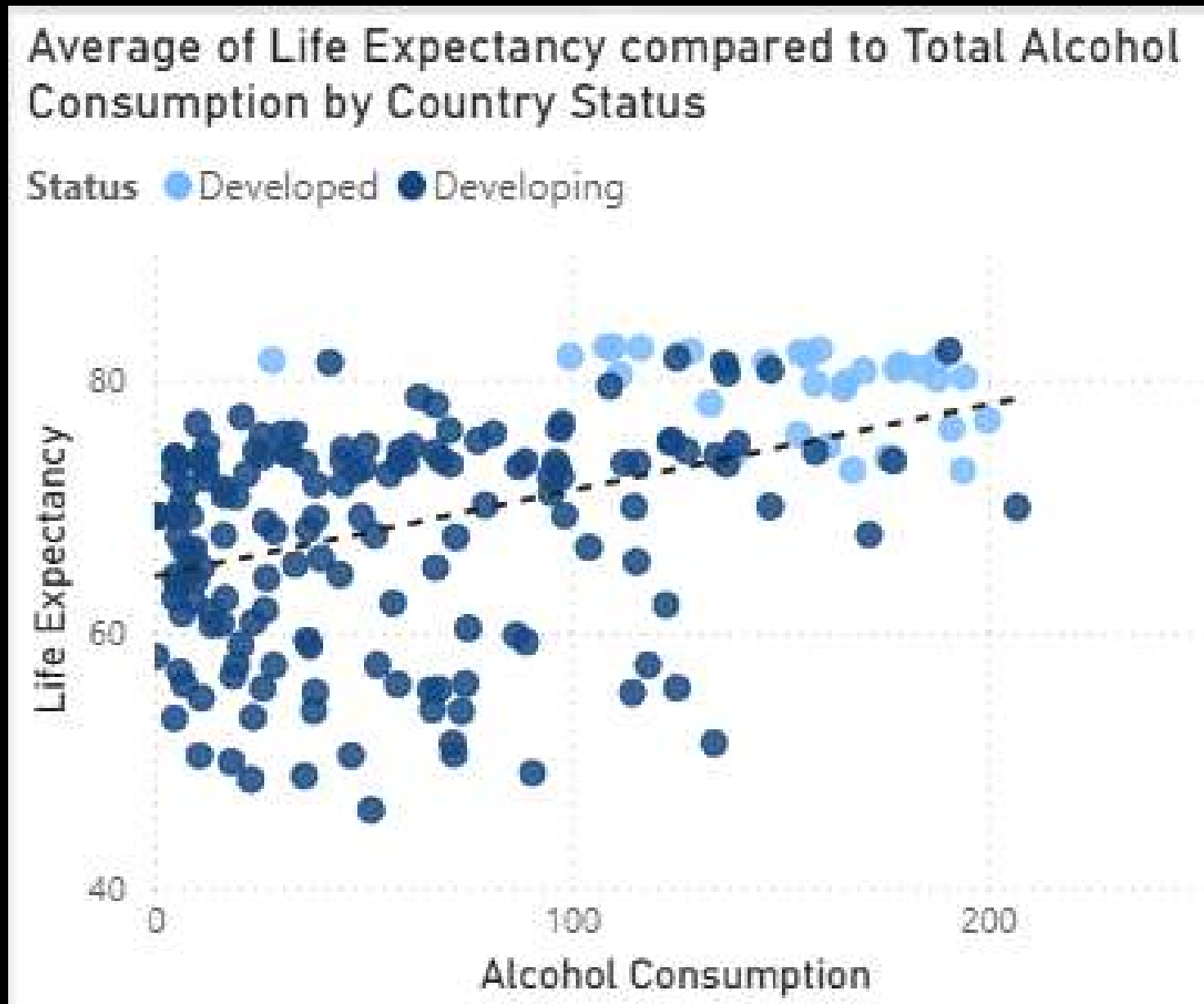


Figure 12. Average of Life Expectancy compared to Total Alcohol Consumption

- Slight **positive** correlation overall.
→ Higher alcohol consumption tends to have higher life expectancy on average.
- Developed countries clusters at higher life expectancy levels across a broad alcohol range.



May reflect **income/development level**, rather than a direct relationship with life expectancy

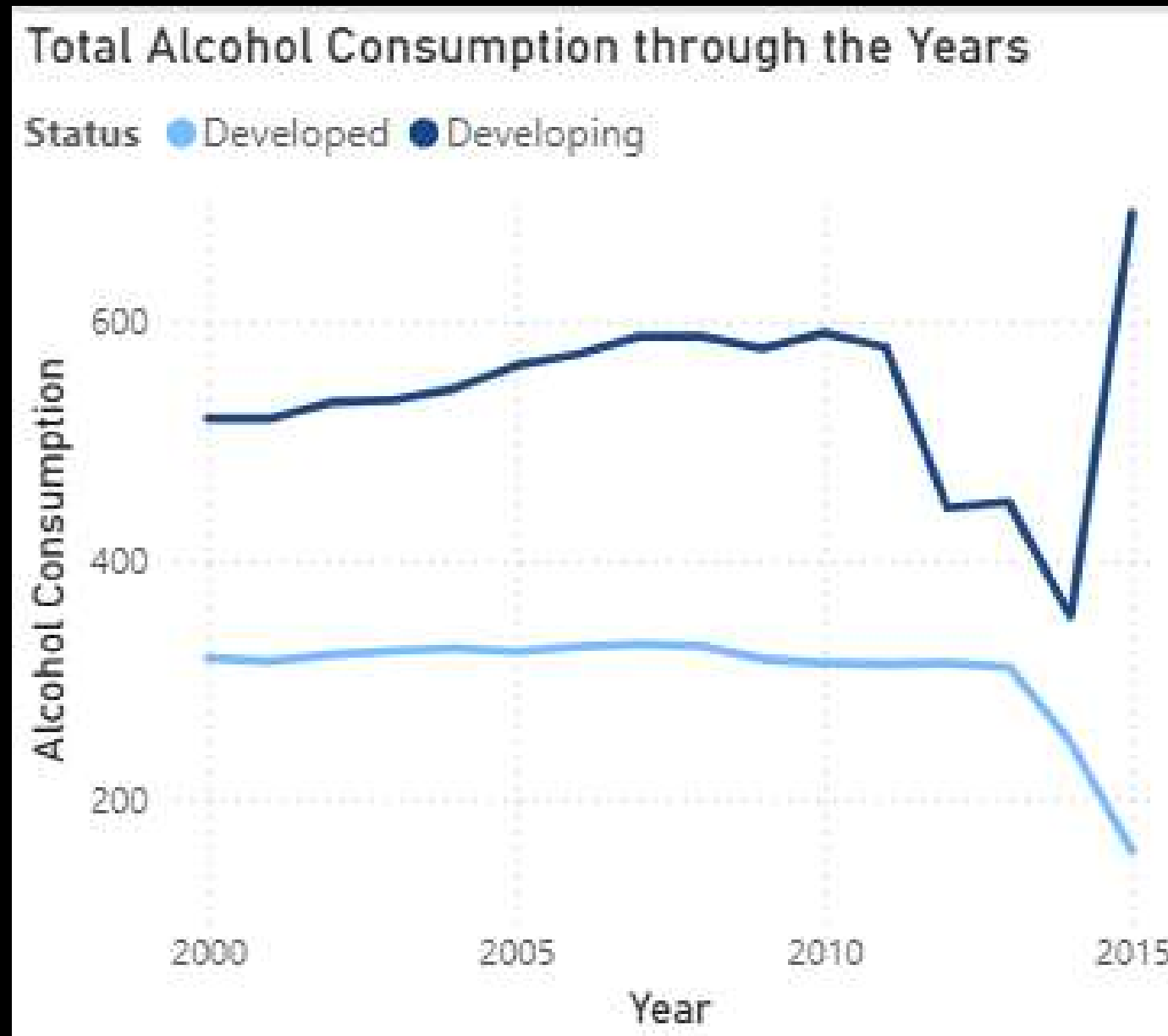


Figure 13. Total Alcohol Consumption through the Years

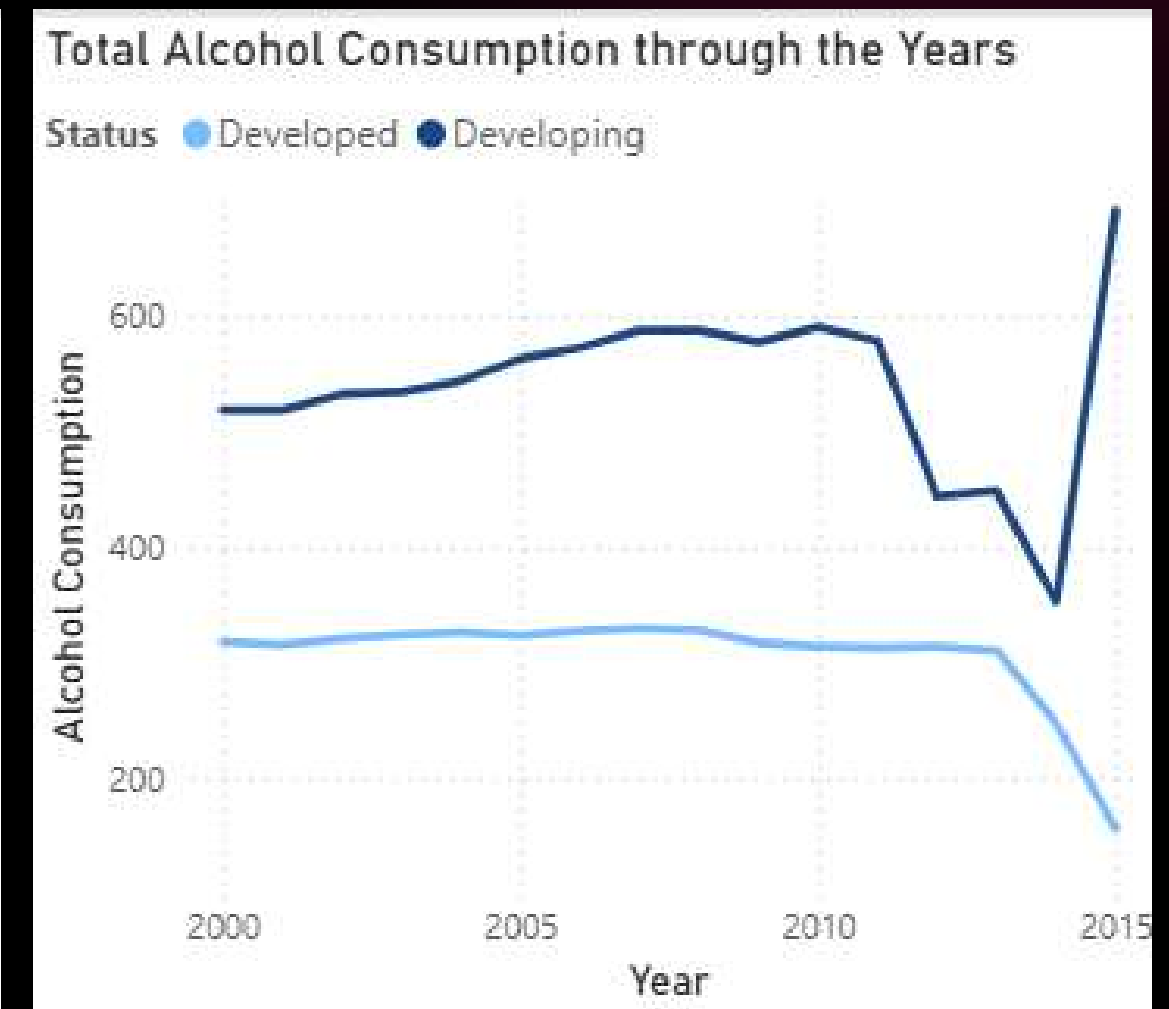
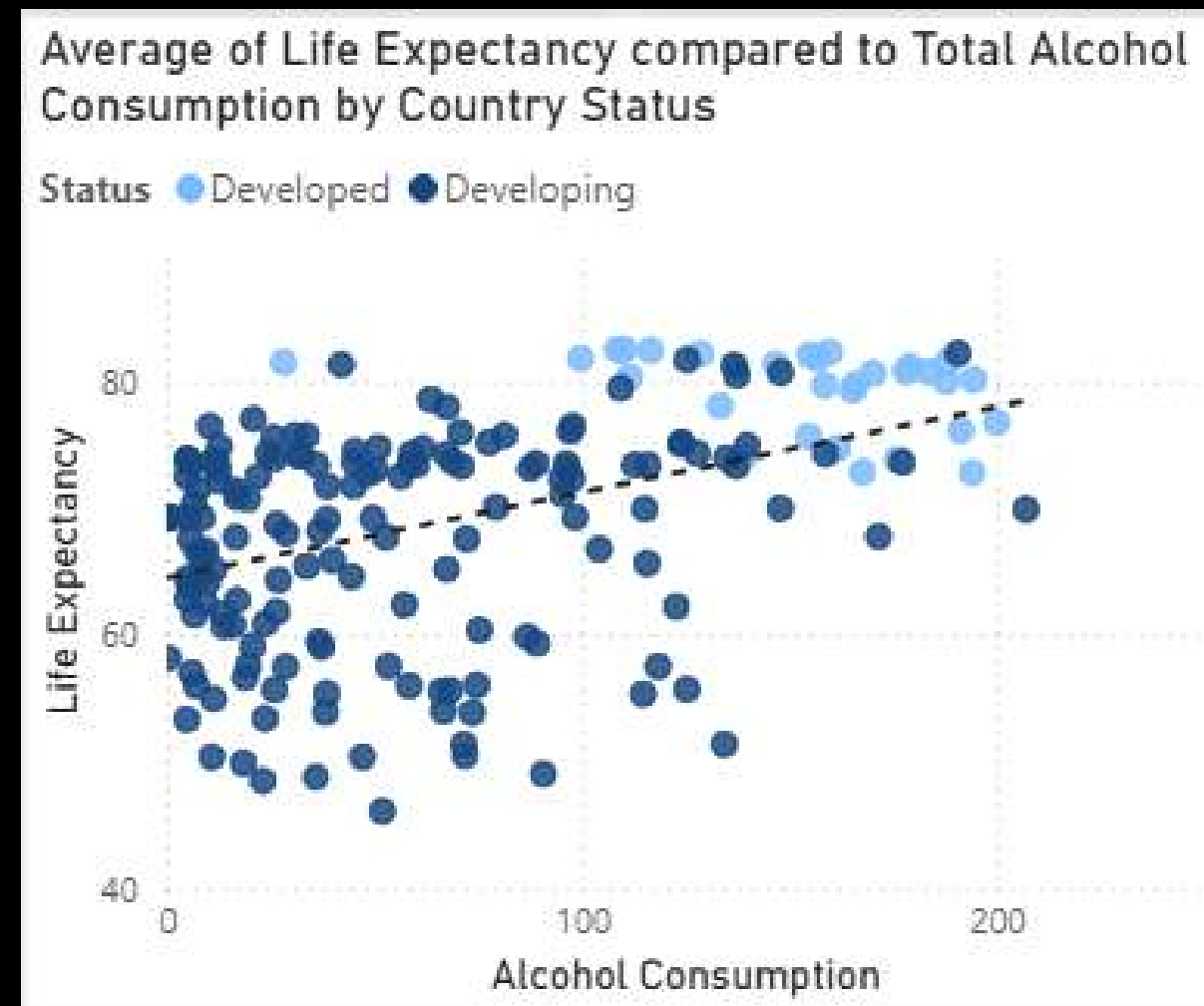
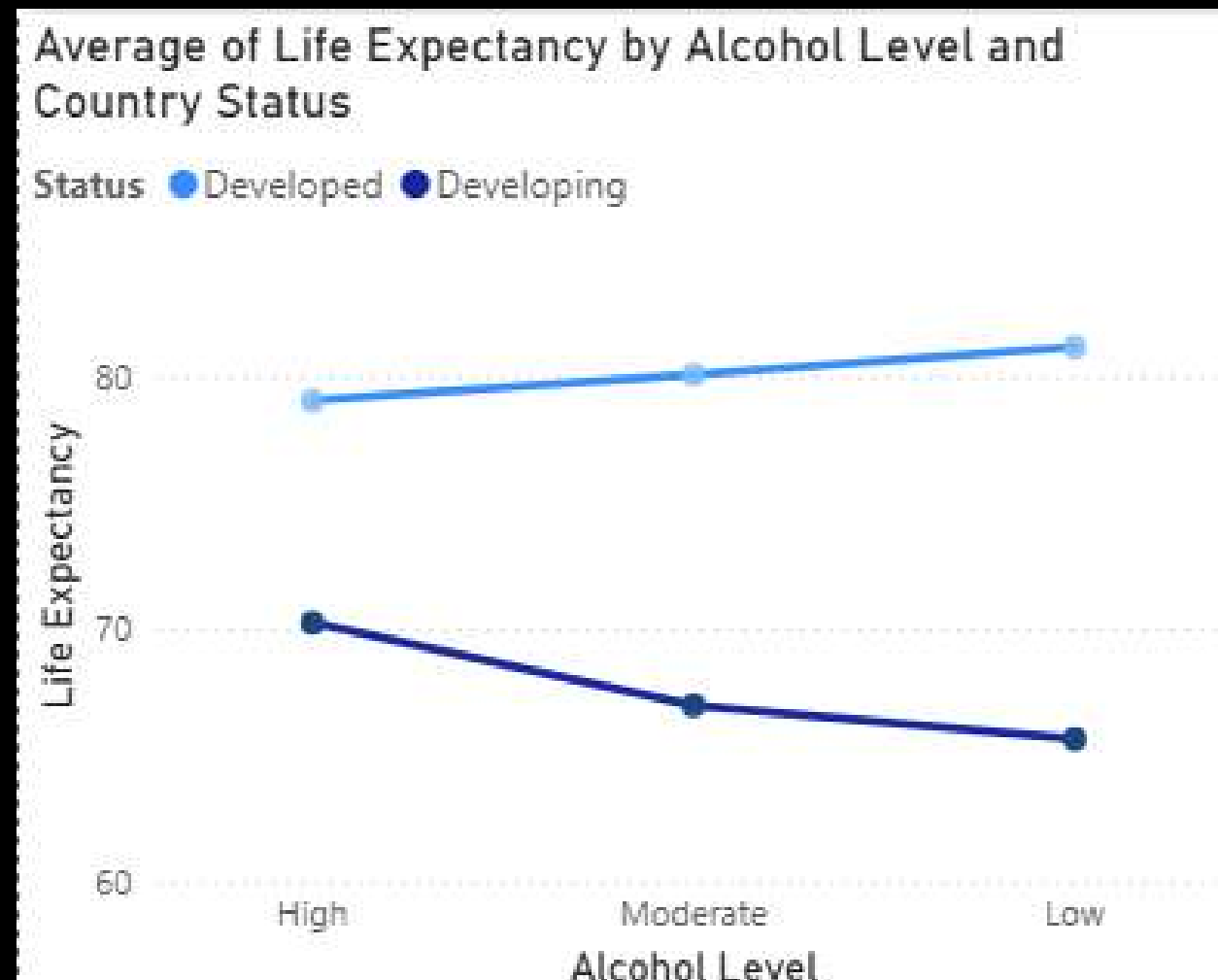
- **Developed countries:** stable over time.
- **Developing countries:** higher and more fluctuating.
- Life Expectancy differences remain.



Developed countries consistently higher



Figure 11, 12, 13.



Alcohol consumption does correlates with life expectancy, but this likely **reflects broader development factors**, not direct health benefits of drinking



- Life Expectancy shows the **strongest positive correlation** with economic conditions and resources.
- BMI shows a **non-linear** relationship.
- Lifestyle factors (i.e. alcohol consumption) has a **weak and/or unclear relationship**, reflects more of overall wealth & healthcare access.

Data Limitations

01

Imbalanced sample size:

Developed vs. Developing countries

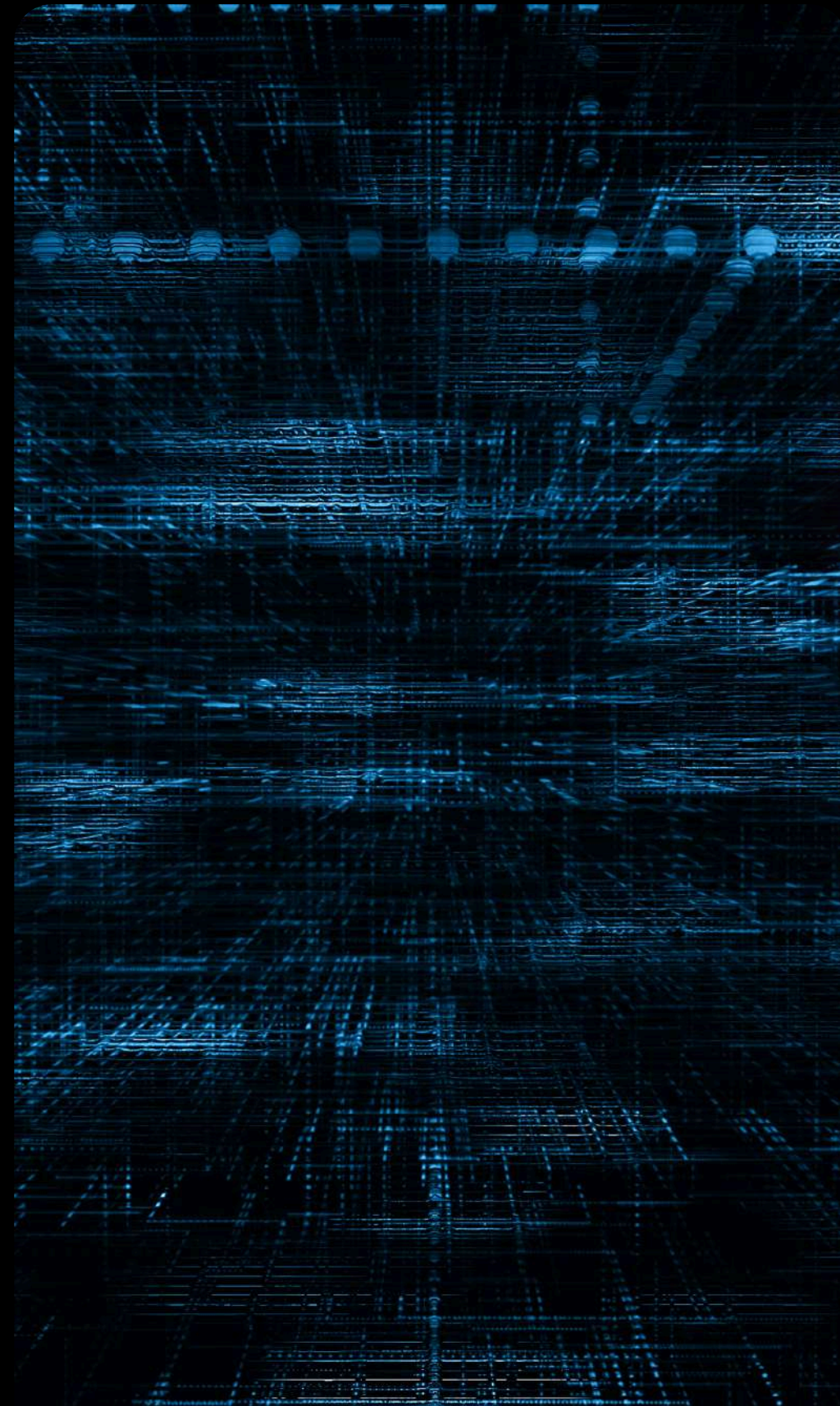
02

Missing data:

Affecting the accuracy of averages
& totals



Results may be biased and less accurate



Conclusion

- Higher immunization coverage is **strongly linked** to higher life expectancy, especially in developing countries.
- Income and healthcare access are the **biggest drivers** of longer life expectancy.
- Lifestyle factors (alcohol, BMI, etc.) have **weaker or mixed correlations** and often reflect overall development levels more than direct impact.

Thank You

FOR YOUR ATTENTION

References

- [1] <https://www.kaggle.com/datasets/kumarajarshi/life-expectancy-who>
- [2] <https://www.nhlbi.nih.gov/calculate-your-bmi>